Experiment 4: Word Frequency Analysis and Word Cloud

```
!pip -q install wordcloud matplotlib

import re
from collections import Counter
import matplotlib.pyplot as plt
from wordcloud import WordCloud
```

```
text = """
Natural language processing (NLP) is a subfield of artificial
intelligence. NLP deals with the interaction between computers
and human language. It helps computers understand, interpret
and generate human language effectively.
"""

# Convert to lowercase and remove punctuation
words = re.findall(r"\b[a-zA-Z0-9]+\b", text.lower())

# Count all tokens and word frequencies
total_words = len(words)
frequency = Counter(words)

# Identify every word tied for the highest frequency
maximum_frequency = max(frequency.values())
most_frequent_words = [
    word for word, count in frequency.items()
    if count == maximum_frequency
]

print("Total number of words:", total_words)
print("Most frequent word(s):", most_frequent_words)
print("Maximum frequency:", maximum_frequency)
print("\nWord Frequencies:")

for word, count in frequency.most_common():
    print(f"{word}: {count}")
```

```
Total number of words: 30
Most frequent word(s): ['language']
Maximum frequency: 3

Word Frequencies:
language: 3
nlp: 2
computers: 2
and: 2
human: 2
natural: 1
```

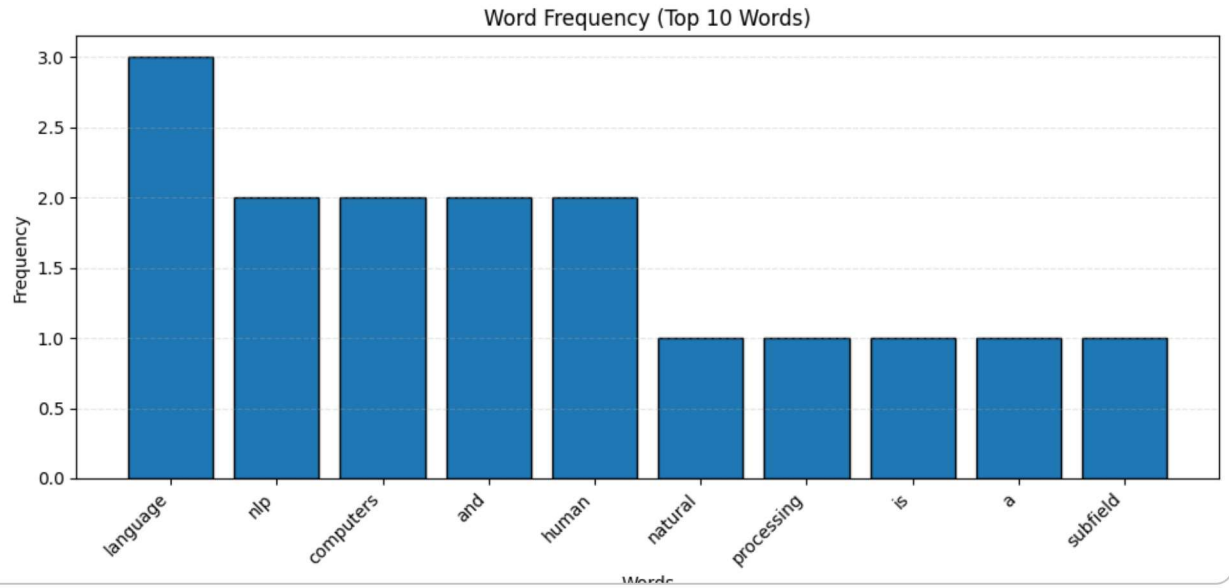
```
processing: 1
is: 1
a: 1
subfield: 1
of: 1
artificial: 1
intelligence: 1
deals: 1
with: 1
the: 1
interaction: 1
between: 1
it: 1
helps: 1
understand: 1
interpret: 1
generate: 1
effectively: 1
```

```
top10 = frequency.most_common(10)

labels = [word for word, count in top10]
counts = [count for word, count in top10]

plt.figure(figsize=(10, 5))
plt.bar(labels, counts, edgecolor="black")

plt.title("Word Frequency (Top 10 Words)")
plt.xlabel("Words")
plt.ylabel("Frequency")
plt.xticks(rotation=45, ha="right")
plt.grid(axis="y", linestyle="--", alpha=0.3)
plt.tight_layout()
plt.show()
```




```
wordcloud = WordCloud(
```

```
    width=1000,
    height=500,
    background_color="white",
    collocations=False
).generate(" ".join(words))

plt.figure(figsize=(12, 6))
plt.imshow(wordcloud, interpolation="bilinear")
plt.axis("off")
plt.title("Word Cloud Representation")
plt.tight_layout()
plt.show()
```

